# Providing reliable space heating and pre-heating of domestic hot water for small

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

## Executive Summary

Generated

| | |
|---|---|
| Project ID | **a_compact_commercial_airsource_heat_pump** |
| Industry Domain | **commercial_hvac** |
| Engine Version | **PDF Engine v2.1 (Full Detail)** |
| Target Quantity | **Not specified** |

### ARCHITECTURE AT A GLANCE

| MODULES | BILL OF MATERIALS ROWS | IDENTIFIED RISKS |
|---|---|---|
| **9** | **0** | **27** |

### ECONOMICS

| ESTIMATED UNIT COST | TARGET CEILING | HEADROOM |
|---|---|---|
| **£0.00** | **£4,500.00** | **£4,500.00** |

| TOTAL NON-RECURRING ENGINEERING | UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING |
|---|---|
| **£0.00** | **£0.00** |

# Contents

# 1. Brief & Requirements

## 1.1 Overview & Context

**Subject / Use Case**

Providing reliable space heating and pre-heating of domestic hot water for small commercial properties through an efficient hydronic system that keeps flammable refrigerants safely outdoors.

**Mission Statement**

To accelerate the decarbonization of UK small commercial buildings by providing an affordable, highly efficient, and low-GWP R290 hydronic split heat pump solution.

**Target Customers**

Small office building owners, boutique hotels, retail spaces, light industrial facilities, and UK HVAC installers specializing in light commercial retrofits.

**Why Now (Market Timing)**

Tightening UK F-Gas regulations are rapidly phasing out traditional highly polluting refrigerants, while government grants and net-zero targets create unprecedented demand for affordable, MCS-certified heating solutions.

### Full Synthesis Report

The UK commercial heating sector is currently undergoing a massive transformation, heavily driven by aggressive government decarbonization targets and the phasing down of high-GWP HFC refrigerants under F-Gas regulations. Developing a 30kW air-source heat pump specifically engineered for small commercial buildings and using R290 (propane) offers a prominent strategic advantage. R290 has a negligible Global Warming Potential (GWP of 3), thereby future-proofing new installations against tightening environmental legislation. Achieving a COP of at least 3.5 at A2/W35 validates a high operational efficiency that is ideally suited for the UK's temperate, damp climate. The proposed hydronic split architecture—which confines the highly flammable R290 charge strictly to the outdoor unit while circulating heated water to the indoor hydro-box—dramatically mitigates indoor safety risks and avoids complex indoor ATEX compliance procedures. At an ambitious target unit cost of £4,500 at production volumes of 500 units per year, the product positions

**Data Sources:**

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

itself very competitively for light commercial retrofits, such as boutique hotels, small offices, and retail spaces. Attaining MCS (Microgeneration Certification Scheme) certification is an absolute necessity in this region, as it unlocks vital UK government incentives and commercial grants for the end customer. Mechanically, the integration of a commercial-grade stainless steel heat exchanger, an R290-optimized scroll compressor, and robust axial fans will require diligent supply chain sourcing and strong negotiation to maintain the strict £4,500 manufacturing cost limit. Subsequent engineering efforts must also heavily focus on optimized defrosting cycles, preventing freezing in the external water loop during power outages, and superior acoustic dampening to easily comply with restrictive urban noise ordinances.

**Data Sources:**

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

## 1.2 Market & Competitors

### Competitor Landscape

#### Ideal Heating

| | |
|---|---|
| Product | **ECOMOD 32kW ASHP** |
| Pricing Strategy | **£6,500 - £7,500 trade price** |
| Technical Specs | **32kW nominal output, R32 refrigerant, COP ~3.4 at A2/W35, monobloc architecture.** |

**STRENGTHS**

Strong UK distribution network, highly recognized commercial boiler brand, straightforward monobloc installation.

**WEAKNESSES**

Uses R32 which is classified as an HFC and subject to F-Gas phase-downs; higher GWP than R290.

**DIFFERENTIATION ANGLE**

Use R290 instead of R32 to future-proof the installation, while keeping the price £2,000 lower through optimized 500-unit/year volume scaling.

#### Vaillant

| | |
|---|---|
| Product | **aroTHERM plus (Cascade System)** |
| Pricing Strategy | **£8,000+ (for 2-3 cascaded units to reach 30kW)** |
| Technical Specs | **Uses multiple 12kW/15kW R290 units cascaded to reach 30kW. COP 3.6 at A2/W35.** |

**STRENGTHS**

Excellent R290 pedigree, exceptionally quiet axial fans, massive UK market share, easy MCS compliance.

**WEAKNESSES**

Requires cascading multiple smaller residential units for 30kW, increasing footprint, piping complexity, and total cost.

**DIFFERENTIATION ANGLE**

Provide a single, purpose-built 30kW commercial chassis to minimize footprint and halve the total hardware cost compared to cascading residential units.

**Data Sources:**

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

## Mitsubishi Electric

| | |
|---|---|
| Product | **Ecodan CAHV P500** |
| Pricing Strategy | **£12,000+** |
| Technical Specs | **43kW output, R407C (older) or R32 variations, scroll compressors, commercial hydronic.** |

#### STRENGTHS

Ultra-reliable scroll compressor technology, heavy commercial durability, built-in redundancy.

#### WEAKNESSES

Significantly oversized and overpriced for the small 30kW commercial segment; uses older or higher-GWP refrigerants.

#### DIFFERENTIATION ANGLE

Target the exact 30kW sweet spot for light commercial use with a much lower £4,500 BOM, focusing specifically on ultra-low GWP R290.

**Data Sources:**

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

# 1.3 Engineering Constraints

## Core Targets

| | |
|---|---|
| Unit Cost Ceiling | **£4,500.00** |
| Max Mass | **-** |
| Target Process | |
| Target Material | |
| Tolerance Target | |
| Production Quantity | |
| Batch Size | **500** |

## Research Sources

| TITLE | TYPE | YEAR | RELEVANCE |
|---|---|---|---|
| **MCS Heat Pump Standard (MIS 3005)** | Regulatory Standard | 2026 | Crucial scheme for the UK market eligibility for government grants; details strict efficiency, sizing, and acoustic requirements. |
| **UK F-Gas Regulation Guidelines** | Government Policy | 2026 | Validates the strategic selection of R290 and dictates long-term phase-down schedules for alternative HFC refrigerants. |
| **BSRIA UK Heat Pump Market Report** | Market Research | 2026 | Provides necessary market sizing, volume expectations, and pricing dynamics for 30kW light commercial systems in the UK. |
| **Danfoss R290 Compressor Application Guide** | Technical Application Note | 2026 | Informs optimal scroll compressor integration, lubrication requirements, and ATEX safety directives for R290 handling. |
| **EN 378: Refrigerating systems and heat pumps** | Engineering Standard | 2026 | Dictates maximum charge limits, safety measures, and outdoor ventilation design criteria for handling A3 (highly flammable) refrigerants. |

**Data Sources:**
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

# 2.1 MIS_3005

## MCS Installation Standard for Heat Pumps

**Status: not-started** Owner: Compliance Engineer

### Standard Summary

Sets the requirements for contractors undertaking the design, supply, installation, and commissioning of heat pump systems in the UK.

### Applicability Assessment

> Required for the end-user to claim Boiler Upgrade Scheme (BUS) or similar commercial decarbonization grants.

### Engineering Design Impact

Requires product performance data to be tested to EN 14511 and acoustic data to be strictly validated.

**EVIDENCE REQUIRED**

Independent MCS product certification and lab test reports from an accredited facility.

**GAP ACTION**

**Engage a UKAS-accredited testing body to begin MCS product onboarding.**

**Data Sources:**
[LLM] Gemini 3.1 Pro — standards extraction

# 2.2 EN_378

## Safety and Environmental Requirements for Heat Pumps

**Status: not-started** | **Owner: Mechanical Design Lead**

### Standard Summary

Pan-European standard covering the design, construction, installation, and maintenance of refrigerating systems.

### Applicability Assessment

> Directly applies to the outdoor unit containing the flammable R290 (A3 class) refrigerant.

### Engineering Design Impact

Demands isolated electrical compartments, spark-free axial fans, and specific pressure relief mechanisms.

**EVIDENCE REQUIRED**

Design FMEA, ATEX component certificates, and physical leak dispersion testing reports.

**GAP ACTION**

**Audit initial BOM for ATEX-rated electrical components in the outdoor unit.**

**Data Sources:**
[LLM] Gemini 3.1 Pro — standards extraction

# 2.3 EN_14511

## Air Conditioners and Heat Pumps with Electrically Driven Compressors

**Status: not-started** | **Owner:** Thermodynamics Engineer

### Standard Summary

Standard for testing and rating of heat pumps for rating the capacity and COP at specified outdoor and water temperatures.

### Applicability Assessment

> All performance claims, specifically the 30kW output and COP >= 3.5 at A2/W35.

### Engineering Design Impact

Designs must be optimized for exact testing envelopes; requires a highly efficient stainless steel heat exchanger and scroll compressor.

#### EVIDENCE REQUIRED

Calibrated laboratory test reports demonstrating capacity and COP compliance.

#### GAP ACTION

**Build first thermodynamic prototype to test internally at A2/W35 parameters.**

**Data Sources:**
[LLM] Gemini 3.1 Pro — standards extraction

# 2.4 PED_2014_68_EU

## Pressure Equipment Directive / UKCA Equivalent

**Status: not-started** | **Owner: Manufacturing Engineer**

### Standard Summary

Regulates the safety of pressurized equipment, including refrigerant loops in HVAC equipment.

### Applicability Assessment

> Applies to the compressor, copper piping, and stainless steel brazed plate heat exchanger.

### Engineering Design Impact

Wall thicknesses, brazing procedures, and pressure limits of the R290 loop must exceed generic commercial thresholds.

**EVIDENCE REQUIRED**

Material certificates, brazing qualification records, and high-pressure burst test results.

**GAP ACTION**

**Confirm PED category based on the R290 total mass and maximum operating pressure.**

**Data Sources:**

[LLM] Gemini 3.1 Pro — standards extraction

# 2.5 EN_14825

## Testing and Rating at Part Load Conditions

Status: not-started

Owner: Systems Engineer

### Standard Summary

Establishes calculating the Seasonal Coefficient of Performance (SCOP) for space heating.

### Applicability Assessment

> Required for ErP (Energy-related Products) directive labelling and UKCA marking.

### Engineering Design Impact

Requires the integration of variable-speed (inverter) drives on the scroll compressor and fans to maintain efficiency at part loads.

**EVIDENCE REQUIRED**

Detailed SCOP calculation dossier based on part-load testing data.

**GAP ACTION**

**Select an inverter drive appropriately sized for a 30kW scroll compressor.**

**Data Sources:**

[LLM] Gemini 3.1 Pro — standards extraction

# 3. Sizing & Spatial Optimisation

**INFEASIBLE DESIGN ALLOCATION**
The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

## System Envelope

| KIND | RULES DOMAIN | FEASIBLE |
|---|---|---|
| generic_room | commer-cial_hvac | No |

| | |
|---|---|
| Internal Dimensions (W×D×H) | **5000 × 5000 × 3000 mm** |
| Total Internal Floor Area | **25 m²** |
| Total Internal Volume | **75 m³** |
| Available Floor Budget | **21.25 m²** |

## Module Dimension Allocations

| MODULE | W × D × H (MM) | AREA M² | MOUNT | SCALE |
|---|---|---|---|---|
| **compressor_module** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **evaporator_module** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **condenser_module** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **refrigerant_circuit** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **air_handling_unit** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |

**Data Sources:**
[Deterministic] Sizing solver and envelope constraints

| | | | | |
|---|---|---|---|---|
| **power_electronics** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **main_controller** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **structural_chassis** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |
| **indoor_hydrobox** | 2236 × 2236 × 2400 | 5 | floor | mass_pro-portional |

### Identified Conflicts

- Total module footprint (45.0m2) exceeds envelope budget (21.3m2)

### Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

**Data Sources:**
[Deterministic] Sizing solver and envelope constraints

# 4.1 Compressor & Isolation Assembly

Lead Time: 16 weeks | Status: draft

**Module Purpose**

Compresses low-pressure R290 vapor into a high-pressure, high-temperature state to drive the heating cycle.

**Why it matters to the system**

> It is the direct driver of the system's 30kW capacity and efficiency. Without a high-grade, R290-certified compressor, achieving the required COP of 3.5 at A2/W35 and maintaining the £4,500 target cost is impossible.

**Detailed Technical Description**

The R290 scroll compressor assembly acts as the thermodynamic engine of the heat pump. Propane (R290) has exceptional volumetric capacity and low discharge temperatures compared to traditional HFCs, allowing the compressor to effectively and efficiently reach high output temperatures without extreme operational strain. This unit employs an inverter-driven variable speed mechanism designed to maximize the Seasonal Coefficient of Performance (SCOP) by adhering closely to part-load energy demands.

Due to the A3 flammability classification of propane, the compressor utilizes specialized seals and non-sparking electrical terminals. Additionally, R290 has a high solubility with standard compressor lubricants, meaning the integration of a tailored crankcase heater and specialized POE (polyolester) oil is critical to prevent oil starvation at start-up. The entire chassis is suspended on heavy-duty elastomeric isolators to curb low-frequency vibrations, which is vital for passing the stringent acoustic requirements of the UK MCS standard.

**SYSTEM INPUTS**

**SYSTEM OUTPUTS**

- High-pressure high-temperature R290 vapor
- Acoustic/vibration emissions

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

- Low-pressure low-temperature R290 vapor
- 3-Phase AC Power (Modulated by Inverter)
- Lubricating oil

**Data Sources:**

[LLM] Gemini 3.1 Pro — module decomposition

## 4.1.2 Specifications & Risk Profile

### Expected Key Parts

- R290-optimized Scroll Compressor
- Vibration isolation mounts
- Acoustic damping jacket
- Crankcase heater

#### Identified Failure Modes

- Liquid slugging destroying the scroll sets during sudden defrost cycling
- Oil starvation caused by excessive R290 dilution into the lubricant
- Electrical terminal arcing causing a hazardous localized ignition event
- Thermal overload due to high compression ratios in extreme cold

#### Open Questions

- Final volumetric efficiency degradation at -15C ambient
- Long-term wear rates of seals exposed strictly to R290

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.1.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.2 Air-Side Evaporator Coil Assembly

Lead Time: 12 weeks

Status: draft

### Module Purpose

Extracts thermal energy from the ambient air to vaporize the liquid R290 refrigerant.

### Why it matters to the system

> Its physical design dictates the system's ability to pull heat in cold environments and directly controls the frequency of defrost cycles, which negatively impact SCOP.

### Detailed Technical Description

The air-side evaporator coil is a massive heat exchanger responsible for capturing latent and sensible heat from the outdoor air. It utilizes a grid of copper tubes mechanically expanded into aluminum fins. To combat the damp, often near-freezing conditions typical of the UK winter, the fins are coated with a hydrophilic material. This coating minimizes droplet surface tension, accelerating moisture runoff and delaying the detrimental accumulation of frost.

When frost inevitably builds, insulating the coil and dropping system capacity, the unit initiates a reverse-cycle defrost. A specialized defrost heater cable lines the condensate pan and the bottom row of the coil to prevent re-freezing of the drained water. Designing this module primarily requires balancing the fin density (FPI). Too dense, and the coil blocks rapidly with ice; too sparse, and the heat pump fails to achieve the 30kW target footprint and ErP efficiency ratings.

#### SYSTEM INPUTS

- Ambient air
- Low-pressure R290 liquid-vapor mixture
- Defrost resistance power

#### SYSTEM OUTPUTS

- Cooled ambient air
- Low-pressure R290 vapor
- Condensate runoff

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.2.2 Specifications & Risk Profile

### Expected Key Parts

- Hydrophilic finned copper coil
- Defrost base heater wire
- Condensate collection tray
- Air temperature sensors

#### Identified Failure Modes

- Ice accumulation crushing the lower fin rows
- Micro-leaks developing at the copper tube U-bends due to thermal cycling fatigue
- Condensate drain blockage leading to water pooling and freezing in the base pan

#### Open Questions

- Exact frosting rate in UK high-humidity/low-temp microclimates
- Tolerance stack-up impacts on coil-to-frame vibration

**Data Sources:**

[LLM] Gemini 3.1 Pro — module decomposition

## 4.2.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.3 Water-Side Condenser Assembly

Lead Time: 14 weeks | Status: draft

### Module Purpose

Transfers heat from the hot, pressurized R290 gas directly into the building's hydronic heating loop.

### Why it matters to the system

> It establishes the thermal bridge to the end-user while guaranteeing the absolute segregation of the flammable R290 charge from the building's indoor environment.

### Detailed Technical Description

The water-side condenser assembly provides the critical boundary between the volatile, hazardous outdoor R290 refrigerant circuit and the safe, non-toxic indoor hydronic loop. It primarily consists of a highly efficient Brazed Plate Heat Exchanger (BPHE) manufactured from 316-grade stainless steel. As the hot propane gas flows through the alternating channels of the BPHE, it condenses into a liquid, transferring latent heat through the extremely thin steel plates to the counter-flowing water circuit.

Due to the strict PED (Pressure Equipment Directive) Category II/III requirements associated with flammable Group 1 gases, the structural integrity of the BPHE's brazed joints must withstand burst pressures drastically higher than standard HFC units. The exchanger is tightly encapsulated in closed-cell polyurethane insulation to prevent thermal loss in the outdoor ambient environment and prevent external condensation during summertime cooling cycles (if configured).

#### SYSTEM INPUTS

- High-pressure high-temperature R290 vapor
- Return water from interior circuit

#### SYSTEM OUTPUTS

- High-pressure R290 liquid
- Hot supply water (up to 75C)

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.3.2 Specifications & Risk Profile

### Expected Key Parts

- SS316 Brazed Plate Heat Exchanger (BPHE)
- Polyurethane insulation jacket
- Water flow switch
- Temperature thermistors

#### Identified Failure Modes

- Plate rupture due to water-side freezing during a winter power outage
- Mineral scaling on the water side severely reducing heat transfer
- Internal brazing failure leading to highly flammable R290 injection into the water loop

#### Open Questions

- Water-side pressure drop limits across commercial installer pumps
- Long-term fatigue limits of plates under extreme cycling

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.3.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.4 Refrigerant Routing & Control Module

Lead Time: 10 weeks | Status: draft

### Module Purpose

Manages phase states, volumetric flow rates, and directional routing of the R290 throughout the system.

### Why it matters to the system

> It controls system superheat and subcooling, extracting maximum efficiency on a micro-scale while maintaining the safety boundary of the explosive R290.

### Detailed Technical Description

The refrigerant circuit stitches together all key thermodynamic modules into a functional loop. Its primary active component is the Electronic Expansion Valve (EEV). Controlled by high-speed step motors receiving PID-loop commands from the main control board, the EEV modulates precisely to drop system pressure, creating a highly regulated liquid/vapor mixture that absorbs heat. The unit also houses a 4-way reversing valve, essential for flipping the flow of hot gas backward through the evaporator to perform necessary winter defrosting.

Safety relies heavily on this module's fabrication. Because it circulates A3 highly flammable propane, standard silver-brazing techniques must be elevated to prevent minute leaks. Components like the filter drier ensure that no moisture causes internal acidic buildup. Under EN 378 regulations, pressure transducers constantly monitor the circuit, triggering hard electrical shutdowns far below the mechanical burst limits of the pipework.

#### SYSTEM INPUTS

- Liquid/Vapor R290 from Evaporator/Condenser
- Control signals (5V/12V DC)

#### SYSTEM OUTPUTS

- Precisely metered R290 flows
- System state reversals (Heating/Cooling/Defrost)

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

# 4.4.2 Specifications & Risk Profile

## Expected Key Parts

- Electronic Expansion Valve (EEV)
- 4-Way Reversing Valve
- Filter Drier
- High/Low Pressure Transducers
- Copper Pipework Assembly

### Identified Failure Modes

- EEV stepper motor binding, locking the valve open/closed
- 4-Way valve sticking mid-shift, destroying unit capacity
- Vibration-induced hairline fracturing in the copper capillary tubes leaking R290
- Filter drier saturation leading to moisture-induced internal icing

### Open Questions

- Exact pipe routing geometry to nullify resonant vibration frequencies
- Time delay lag between EEV modulation and temperature sensor detection

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.4.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.5 Fan & Airflow Assembly

Lead Time: 14 weeks | Status: draft

### Module Purpose

Induces large volumes of ambient air across the evaporator coil to facilitate thermal transfer while adhering to strict noise limits.

### Why it matters to the system

> Generates the necessary airflow to hit 30kW capacity, but must do so without violating rigid acoustic zoning laws or ATEX ignition safety requirements.

### Detailed Technical Description

The air handling module contains the high-diameter axial fans necessary to pull outside air through the restrictive fins of the evaporator coil. To meet the aggressive ErP efficiency mandates and variable part-load requirements (SCOP), these fans employ Brushless DC (BLDC) motors connected to an inverter drive. This allows the system to modulate fan speeds from 20% to 100%, optimizing the power consumed dynamically as outdoor temperatures shift.

The most extreme challenge for this module is acoustics. To meet the stringent UK MCS planning requirements for urban and light commercial retrofits, the fan blade geometry requires biomimetic trailing edge designs to break up tonal noise. Additionally, because a leak of R290 could pool around the fan assembly, all fan motors must feature inherently spark-free topologies complying with ATEX atmospheric requirements, preventing the fans from becoming an ignition source.

#### SYSTEM INPUTS

- Ambient air
- Variable DC power (from Inverter)

#### SYSTEM OUTPUTS

- Directed exhaust air
- Aerodynamic and motor noise

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.5.2 Specifications & Risk Profile

### Expected Key Parts

- Brushless DC (BLDC) Axial Fans
- Aerodynamic fan cowlings
- Bird/Safety grilles
- Spark-free motor enclosures

#### Identified Failure Modes

- Bearing seizure due to continual outdoor exposure and inadequate sealing
- Ice buildup on blades creating immense asymmetrical imbalances and vibration
- Resonant frequencies between the fan chop and the cowling creating drone complaints

#### Open Questions

- Real-world snow ingestion characteristics of the external grilles
- Tolerance variations in fan balancing affecting long-term vibration

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

### 4.5.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.6 Inverter & Power Distribution Unit

Lead Time: 18 weeks | Status: draft

**Module Purpose**

Converts grid AC power into highly modulated, efficient variables for the compressor and DC components.

**Why it matters to the system**

> It determines the overall electrical stability, directly governs the SCOP through inverter efficiency, and presents the largest component cost ceiling risk.

**Detailed Technical Description**

The power electronics module transforms standard 400V 3-Phase commercial grid electricity into the highly tailored outputs required by modern HVAC componentry. Generating 30kW of heat output requires approximately 8.5kW to 10kW of electrical input. The core of this module is the Variable Frequency Drive (VFD) which synthesizes a precise sinusoidal waveform to drive the scroll compressor at infinitely variable speeds. This prevents the massively inefficient 'hard switching' of older systems and enables optimal part-load tracking.

Due to the significant electrical noise generated by rapid switching frequencies in the VFD, this unit houses dense EMI (Electromagnetic Interference) filters and line reactors to meet CE/UKCA EMC compliance. Crucially, as the inverter bridge dissipates approximately 3-5% of its load as waste heat, it is mounted to an aggressive aluminum heat sink that protrudes into the condenser fan airstream. To meet EN 378 safely standards for A3 refrigerants, this entire electrical bay must be sealed off from the mechanical / refrigerant compartment to guarantee separation from potential leaks.

**SYSTEM INPUTS**

- 400V 3-Phase AC Grid Power

**SYSTEM OUTPUTS**

- Variable Frequency 3-Phase AC (Compressor)
- Modulated DC (Fans)
- Stable 24V/12V/5V DC (Controls Bus)

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

- Waste Heat

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.6.2 Specifications & Risk Profile

### Expected Key Parts

- Variable Frequency Drive (VFD) Inverter Bridge
- EMI/RFI filter block
- DC Link Capacitors
- Power distribution PCB
- Aluminum heat sink

#### Identified Failure Modes

- IGBT (Insulated Gate Bipolar Transistor) thermal breakdown
- DC link capacitor degradation leading to excessive ripple currents
- EMI filter saturation causing feedback into the commercial building grid
- Condensation inside electrical box causing short circuits

#### Open Questions

- Exact thermal dissipation load required in high-ambient operations
- Impact of UK commercial power grid voltage sags on stable 30kW output

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

### 4.6.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.7 Main Control & Logic PCB

Lead Time: 12 weeks | Status: draft

### Module Purpose

Acts as the system brain, processing sensor inputs, running thermodynamic algorithms, and enforcing critical safety bounds.

### Why it matters to the system

> Hardware is only as good as the software driving it; this board ensures peak thermodynamic efficiency while acting as the absolute fail-safe against explosive hazards.

### Detailed Technical Description

The logic controller is a densely populated printed circuit board housing the proprietary firmware that governs the entire thermodynamic process. It receives constant telemetry from dozens of high-side and low-side temperature and pressure sensors, allowing it to compute exact superheat and subcooling values in real-time. By manipulating the EEV, compressor inverter, and fan speeds, the controller squeezes every possible fraction of efficiency to meet the 3.5 COP baseline.

Importantly, this module manages the sophisticated safety architecture mandated by EN 378 for highly flammable refrigerants. If a sudden pressure drop or dedicated R290 sniffer detects a leak, the control board forcefully isolates the power supply, shutting down the compressor and triggering evacuation fans (if built-in). Furthermore, it possesses the intelligence to manage anti-legionella cycles for commercial hot water tanks and provides Modbus communication for seamless integration into light commercial building management systems (BMS).

#### SYSTEM INPUTS

#### SYSTEM OUTPUTS

- Pulse Width Modulation (PWM) signals to drivers
- Safety relay cutoffs
- Diagnostics and error reporting codes

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

- 5V/12V DC power
- Thermistor and pressure transducer data
- BMS/Thermostat external triggers
- R290 leak detector signs

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.7.2 Specifications & Risk Profile

### Expected Key Parts

- Main Microcontroller PCB
- ATEX-certified safety relay arrays
- Modbus/RS485 communication terminal
- Real-Time Clock (RTC) chip

#### Identified Failure Modes

- Microcontroller latch-up due to electromagnetic interference
- Flash memory corruption stalling control algorithms in deep freeze conditions
- Safety relay welding closed due to unexpected inductive kickback
- Sensor input ghosting causing erratic system behavior

#### Open Questions

- Exact complexity of firmware needed for cascading units in future iterations
- Susceptibility of the board to moisture intrusion during prolonged storage

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.7.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.8 Enclosure & Acoustic Structure

Lead Time: 8 weeks    Status: draft

**Module Purpose**

Provides the structural backbone, environmental shielding, and noise attenuation foundation for the outdoor unit.

**Why it matters to the system**

> It directly solves the MCS acoustic compliance constraints while acting as the passive mechanical safety limit against pooling flammable gases.

**Detailed Technical Description**

The outdoor chassis establishes the physical boundaries and ruggedness necessary to withstand decadal exposure to brutal UK weathering. Because this unit encapsulates a large 30kW payload, the base pan is constructed from heavily reinforced, galvanized steel preventing long-term sagging or twisting which would rupture internal copper lines. The external skin relies on thick powder-coated sheets engineered to withstand standardized salt-spray testing, essential for coastal retrofit viability.

Two critical functions overlap in the chassis design: acoustics and A3 safety. To pass the stringent noise ordinances, the compressor compartment must be lined with heavy, mass-loaded acoustic barriers absorbing low-frequency noise. Unlike older HFC heat pumps, this enclosure cannot be entirely hermetic at the base; R290 is heavier than air, therefore the base pan incorporates specialized, unimpeded louvers. In the event of a catastrophic internal leak, these vents ensure the explosive gas passively drains out of the unit and disperses safely into the ambient atmosphere before reaching the Lower Flammability Limit (LFL).

**SYSTEM INPUTS**

- Environmental rain, snow, and UV
- Internal vibrations and tonal noise

**SYSTEM OUTPUTS**

- Weatherproofed exterior
- Attenuated acoustic envelope

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

- Component weight loading (up to 250kg)
- Directed safe ventilation patterns (for R290)

**Data Sources:**

[LLM] Gemini 3.1 Pro — module decomposition

## 4.8.2 Specifications & Risk Profile

### Expected Key Parts

- Galvanized standard steel base pan
- Powder-coated access panels
- Mass-loaded vinyl acoustic shielding
- R290 safe ventilation grilles

#### Identified Failure Modes

- Base pan corrosion leading to structural collapse
- Panel rattling developing harmonics with the compressor output
- Debris blocking base vents, resulting in localized R290 pooling internally

#### Open Questions

- Exact vibrational node points that will require extra bracing
- Aesthetic trade-offs required to allow sufficient R290 drainage

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.8.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 4.9 Indoor Hydronic Circulation Hub

Lead Time: 10 weeks | Status: draft

**Module Purpose**

Circulates the hot water produced by the outdoor unit into the existing commercial heating infrastructure safely and efficiently.

**Why it matters to the system**

> It radically simplifes installation for the plumber and firmly removes the need for ATEX-level compliance indoors, keeping the product highly marketable.

**Detailed Technical Description**

The indoor hydrobox represents the 'split' conceptual aspect of this monobloc architecture. While the entire R290 refrigeration cycle remains rigidly sealed outside, this indoor module securely manages the water-side distribution within the commercial building. Housing all critical hydronic components—such as the ErP-grade variable-speed circulating pump, hydraulic expansion vessel, and flow switches—it dramatically shrinks the required installation time and eliminates the complex piecemeal plumbing traditionally required by mechanical contractors.

Crucially, this module contains a backup electric immersion heater. In the event of a catastrophic outdoor unit failure or extreme sub-zero cold snaps surpassing the 30kW capability, this heater steps in to prevent building freezes. It also houses magnetic filtration, preventing historic black iron sludge often found in retrofitted commercial radiators from travelling outside and destroying the fragile micro-channels of the outdoor brazed plate heat exchanger.

**SYSTEM INPUTS**

- Heated water (up to 75C) from outdoor unit
- 230V AC Power (Indoor)
- Domestic Hot Water / Heating loop demands

**SYSTEM OUTPUTS**

- Pressurized hot water flow
- System flow rate telemetry to main board

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

**Data Sources:**
[LLM] Gemini 3.1 Pro — module decomposition

## 4.9.2 Specifications & Risk Profile

### Expected Key Parts

- Variable-speed circulation pump
- 12L to 18L Expansion Vessel
- 3kW to 6kW Backup Immersion Heater
- Magnetic dirt filter / separator
- Motorized 3-way diverter valve

#### Identified Failure Modes

- Circulation pump cavitation due to improper system bleeding
- Expansion vessel bladder rupture causing pressure spikes and relief valve dumps
- Immersion heater scale build-up leading to element burnout
- Magnetic filter saturation restricting flow to the outdoor BPHE

#### Open Questions

- Actual pressure drops of older retrofitted commercial heating loops
- Required expansion vessel sizing for varying commercial pipe volumes

**Data Sources:**

[LLM] Gemini 3.1 Pro — module decomposition

## 4.9.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

**Data Sources:**
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

# 5. Bill of Materials

No parts in bill of materials.

# 6. Cost Waterfall & Economics

| UNIT COST | TARGET CEILING | HEADROOM |
|---|---|---|
| **£0.00** | **£4,500.00** | **£4,500.00** |

### Make vs Buy Split

| PURCHASED (BUY) TOTAL | CUSTOM (MAKE) TOTAL | MAKE % |
|---|---|---|
| **£0.00** | **£0.00** | **0%** |

### Per-Module Roll-up

Per-module cost data not available.

### Non-recurring engineering and overheads

| | |
|---|---|
| Total non-recurring engineering (Tooling, Safety Testing, Compliance) | **£0.00** |

**Grand Total (Unit + Non-recurring Engineering)** **£0.00**

**Data Sources:**
[Deterministic] Domain overhead multiplier model

### Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

**Data Sources:**
[Deterministic] Domain overhead multiplier model

## 7.1 Compressor & Isolation Assembly Risks

### Risk 1: Liquid slugging on start-up

**Base Score: S:5 × L:2 = 10**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Install suction accumulator and execute soft-start algorithms

**Owner: Thermodynamics Engineer**

### Risk 2: Crankcase oil dilution

**Base Score: S:4 × L:3 = 12**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Utilize constant-wattage crankcase heater and R290-specific POE oil

**Owner: Mechanical Design Lead**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.1 Compressor & Isolation Assembly Risks

### Risk 3: Acoustic resonance non-compliance

**Base Score: S:3 × L:4 = 12**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Mass-loaded acoustic blankets and validated rubber isolators

**Owner: Compliance Engineer**

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.2 Air-Side Evaporator Coil Assembly Risks

### Risk 4: Severe frost rendering unit inoperable

Base Score: S:4 × L:4 = 16

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Implement algorithmic demand-defrost based on pressure drop, not just time

Owner: Systems Engineer

### Risk 5: Condensate pan freezing solid

Base Score: S:4 × L:3 = 12

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Integrate robust 150W base pan heater tied to ambient thermistor

Owner: Mechanical Design Lead

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.2 Air-Side Evaporator Coil Assembly Risks

### Risk 6: Fin corrosion from airborne coastal salts

Base Score: S:3 × L:2 = 6

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Use factory-applied epoxy/hydrophilic anti-corrosion coating

Owner: Manufacturing Engineer

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.3 Water-Side Condenser Assembly Risks

### Risk 7: R290 leakage into water stream

**Base Score: S:5 × L:1 = 5**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Helium leak testing at factory and strict PED burst verification

Owner: Compliance Engineer

### Risk 8: Water side freezing and rupturing BPHE

**Base Score: S:5 × L:3 = 15**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Require antifreeze/glycol in the manual or include anti-freeze pump cycling logic

Owner: Systems Engineer

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.3 Water-Side Condenser Assembly Risks

### Risk 9: Poor heat transfer causing low COP

**Base Score: S:4 × L:2 = 8**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Oversize the BPHE surface area by 15% margin

**Owner: Thermodynamics Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.4 Refrigerant Routing & Control Module Risks

### Risk 10: Vibration-induced copper tube fracture (R290 leak)

**Base Score: S:5 × L:3 = 15**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

FEA on pipework and extensive 3-axis vibration testing

**Owner: Mechanical Design Lead**

### Risk 11: 4-Way valve sticking during defrost

**Base Score: S:4 × L:2 = 8**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Select premium pilot-operated valves and test under max pressure differential

**Owner: Thermodynamics Engineer**

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.4 Refrigerant Routing & Control Module Risks

### Risk 12: EEV stepping failure

**Base Score: S:3 × L:2 = 6**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Firmware watchdogs tracking superheat anomalies to trigger safe shutdown

**Owner: Systems Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.5 Fan & Airflow Assembly Risks

### Risk 13: Sparking during theoretical R290 leak pool

**Base Score: S:5 × L:1 = 5**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Mandate strictly non-spark motor topologies and IP67 sealing

**Owner: Compliance Engineer**

### Risk 14: Excessive tonal noise failing MCS acoustic limits

**Base Score: S:4 × L:4 = 16**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Use swept-blade aerodynamic profiles and resilient motor mounts

**Owner: Mechanical Design Lead**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.5 Fan & Airflow Assembly Risks

### Risk 15: Ice forming on fan blades causing imbalance

**Base Score: S:3 × L:3 = 9**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Program periodic high-spin cycles in cold weather to shed ice

**Owner: Systems Engineer**

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.6 Inverter & Power Distribution Unit Risks

### Risk 16: R290 infiltration into electrical bay causing explosion

**Base Score: S:5 × L:1 = 5**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Positively pressurize or physically compartmentalize the electrical housing enclosure

**Owner: Mechanical Design Lead**

### Risk 17: IGBT thermal runaway destroying inverter

**Base Score: S:4 × L:3 = 12**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Direct heat sink into primary airstream and add thermal limit cutoffs

**Owner: Thermodynamics Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.6 Inverter & Power Distribution Unit Risks

### Risk 18: Grid harmonic distortion failing CE tests

**Base Score: S:3 × L:3 = 9**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Ensure properly rated DC chokes and robust EMI filter selection

**Owner: Compliance Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.7 Main Control & Logic PCB Risks

### Risk 19: Failure to trigger shutdown during major R290 leak

**Base Score: S:5 × L:1 = 5**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Implement redundant parallel ATEX safety relays overriding software loops

Owner: Systems Engineer

### Risk 20: Defrost algorithm failing in unique micro-climates

**Base Score: S:4 × L:4 = 16**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Include adaptive AI-based defrost tuning referencing historical weather data

Owner: Thermodynamics Engineer

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.7 Main Control & Logic PCB Risks

### Risk 21: Loss of connectivity to commercial BMS

**Base Score: S:2 × L:3 = 6**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Use robust, optically isolated RS485 transceivers for Modbus lines

**Owner: Systems Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.8 Enclosure & Acoustic Structure Risks

### Risk 22: R290 pooling in lower chassis upon leak

**Base Score: S:5 × L:2 = 10**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Design sloped base pan with oversized, permanently open drain/vent ports

**Owner: Mechanical Design Lead**

### Risk 23: Acoustic panels trapping moisture and causing rust

**Base Score: S:3 × L:4 = 12**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Use closed-cell hydrophobic acoustic foam and ensure adequate air gaps

**Owner: Manufacturing Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.8 Enclosure & Acoustic Structure Risks

### Risk 24: Chassis warp during transport causing pipe stress

**Base Score: S:4 × L:2 = 8**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Reinforce corner pillars and mandate specific forklift lifting geometry

**Owner: Manufacturing Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.9 Indoor Hydronic Circulation Hub Risks

### Risk 25: Pump failure leading to outdoor BPHE freezing

Base Score: S:5 × L:2 = 10

EXISTING CONTROLS

None

PLANNED MITIGATION

Hardwired flow switch interlock that halts outdoor compressor instantly

Owner: Systems Engineer

### Risk 26: Commercial sludge ruining outdoor heat exchanger

Base Score: S:4 × L:4 = 16

EXISTING CONTROLS

None

PLANNED MITIGATION

Integrate premium magnetic filter with easy maintenance access valve

Owner: Mechanical Design Lead

**Data Sources:**
[LLM] Gemini 3.1 Pro — FMEA analysis

## 7.9 Indoor Hydronic Circulation Hub Risks

### Risk 27: Expansion vessel severely undersized for large capacity pipes

**Base Score: S:3 × L:3 = 9**

**EXISTING CONTROLS**

None

**PLANNED MITIGATION**

Allow auxiliary port for installers to easily plumb supplementary expansion tanks

**Owner: Manufacturing Engineer**

**Data Sources:**

[LLM] Gemini 3.1 Pro — FMEA analysis

# 8. Supplier Shortlist

No supplier matches found during the search phase.

# 9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

### Pipeline Execution Summary

| | |
|---|---|
| Project Identifier | **a_compact_commercial_airsource_heat_pump** |
| Research Result | **Generated** |
| System Modules | **9 Modules** |
| Bill of Materials Lines | **0 Parts** |
| Cost Analysis | **Pending** |
| Specialist Reviews | **0 Completed** |
| Supplier Shortlists | **0 Matched** |

# 10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

| SECTION | SOURCE | PROVIDER / SYSTEM | DETAIL |
| --- | --- | --- | --- |
| **Research** | LLM | OpenRouter | Gemini 3.1 Pro via OpenRouter |
| **Research** | USER | user | Founder brief text |
| **Regulatory** | LLM | Unknown LLM | Gemini 3.1 Pro — standards extraction |
| **Modules** | LLM | Unknown LLM | Gemini 3.1 Pro — module decomposition |